

GenAI Training

3-Month (12-Week) Program Overview

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Program at a Glance

Build a strong Generative AI foundation and deliver an LLM-based capstone project in 12 weeks.

DURATION

3 months (12 weeks)

CLASS TIMING

2 hours per day

ASSESSMENTS

Every Friday (weekly)

TOOLING STACK

Python

Transformers

Hugging Face

OpenAI API

LangChain

Prompt Engineering

Streamlit

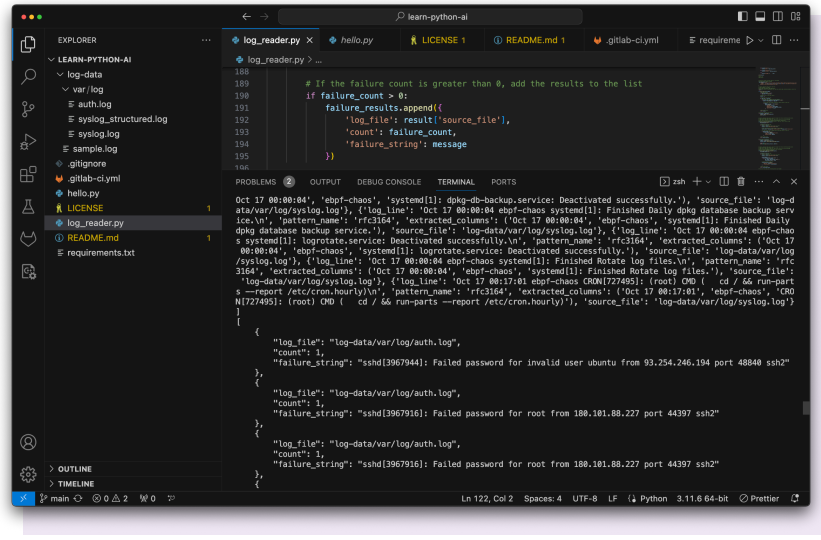
Month 1 — Getting Started (Weeks 1-2)

Week 1: Introduction to Generative AI

- What GenAI is; how it differs from AI/ML/DL
- Common applications and where LLMs fit
- Ethics and responsible use (high-level)

Week 2: Python & API Fundamentals

- Python refresh; working with APIs and JSON
- Virtual environments; Git basics



LEARNING FLOW



Concepts



Setup



First API call



Version control

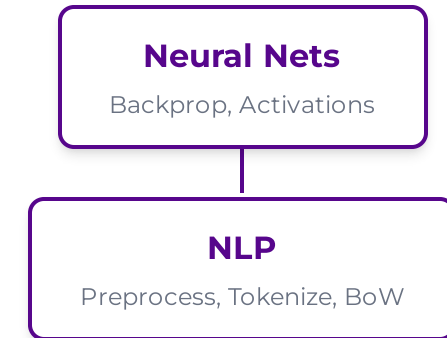
Month 1 — Core Foundations (Weeks 3–4)

Week 3: Deep Learning Basics

- Neural network fundamentals
- Activation functions and backpropagation (concept)
- Probability basics for AI

Week 4: NLP Foundations

- Text preprocessing and tokenization
- Bag of Words baseline intuition



Month 2 - LLMs and Prompting (Weeks 5-6)

Week 5: Transformers & LLMs

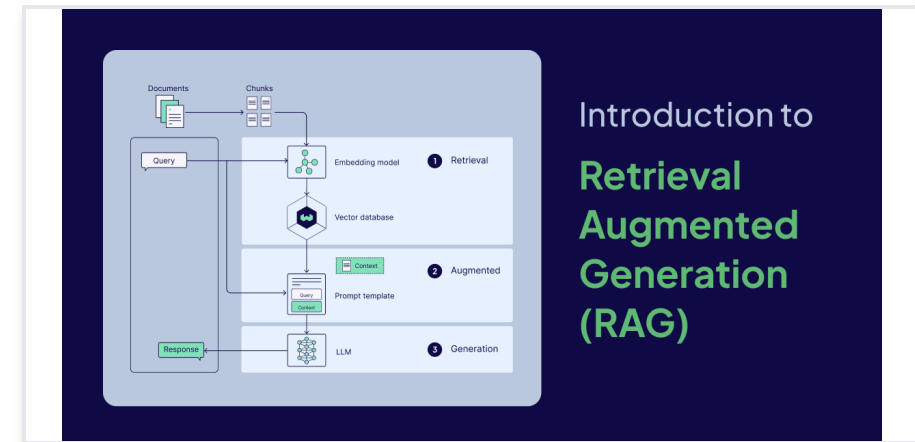
Month 2 - Building with APIs and RAG (Weeks 7-8)

Week 7: OpenAI API Integration

- Text generation and chat models
- Temperature and tokens (conceptual levers)
- Handling API responses

Week 8: LangChain & RAG Introduction

- Embeddings concept; vector database overview
- Retrieval-Augmented Generation (RAG)
- Basic chatbot development



Document

Chunk

Embed

Retrieve

Prompt + Context

Answer

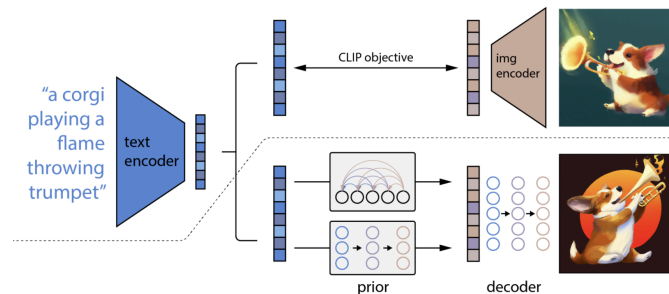
Month 3 - Applied GenAI (Weeks 9-10)

Week 9: Generative AI Applications

- Content generation
- Code generation
- Resume generation
- Automation use cases

Week 10: Image Generation Concepts

- Diffusion models overview
- DALL-E / Stable Diffusion introduction
- Image prompt techniques



Month 3 - Build Improve Deliver (Weeks 11-12)

Week 11: Advanced RAG & Chatbots

- ✓ Document Q&A systems
- ✓ Custom knowledge chatbot
- ✓ Fine-tuning overview (concept only)

Week 12: Capstone Project & Career Preparation

- ✓ Project development and weekly milestones
- ✓ Deployment using Streamlit
- ✓ Resume building and interview prep

Idea →

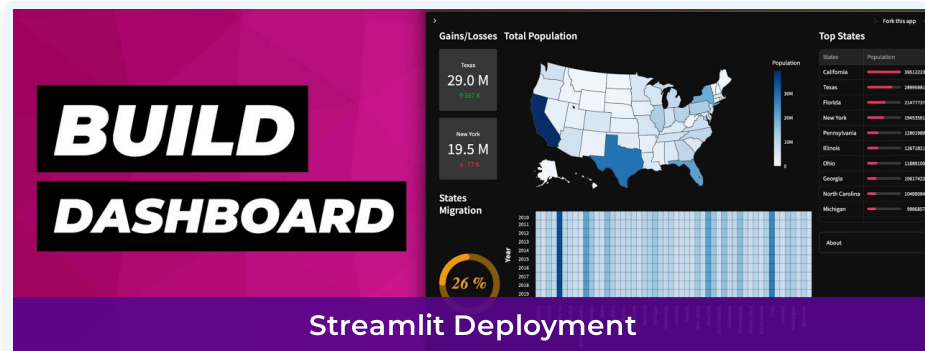
Data/docs →

Build →

Evaluate →

Deploy →

Demo/Viva



Assessments & Student Deliverables

- Weekly assessments (every Friday)
- Prompt engineering assignments
- API integration projects
- One Generative AI capstone project (+ viva)
- GitHub repository (all work versioned)

Assessment Checklist

Use this checklist as a good practice guide for assessment design, evaluation and renewal.

Alignment of Assessment with Course and Program Aims	Yes	No	Unsure
1. Assessment is an integral part of my course planning.	☐	☐	☐
2. Course assessment contributes to overall program goals/aims/objectives.	☐	☐	☐
3. Assessment enables students to demonstrate forms of learning described in all course learning objectives.	☐	☐	☐
4. Assessment contributes to the development of disciplinary mastery.	☐	☐	☐
5. Assessment contributes to the development of other program level goals including accreditation standards and graduate skills or attributes.	☐	☐	☐

Assessment of Learning	Yes	No	Unsure
1. Assessment judges student performance against explicit, relevant criteria and standards.	☐	☐	☐
2. Assessment task is appropriate, given students' prior learning in my discipline.	☐	☐	☐
3. Examiner expectations about performance standards for specific assessment tasks are made transparent for students through the use of marking guidelines, assessment instructions and/or marking rubrics.	☐	☐	☐
4. Assessment tasks meet all course learning objectives, both disciplinary content and relevant graduate skills where appropriate.	☐	☐	☐
5. There is a moderation process for reaching consensus amongst all teaching team members about expected standards of student performance in assessment.	☐	☐	☐
6. The first assessment item occurs early enough in the semester to engage students with their course.	☐	☐	☐
7. The first assessment item is a manageable task designed to build students' skills, knowledge and confidence.	☐	☐	☐
8. Assessment tasks contribute to the development of the graduate skills stated in the course learning objectives, if appropriate.	☐	☐	☐
9. Assessment tasks target coherent, integrated forms of learning rather than disconnected fragments.	☐	☐	☐
10. Assessment tasks engage students in active, authentic learning experiences.	☐	☐	☐



Weekly assessments



Assignments/projects



Capstone + Viva



Repo submission

Outcomes and Next Steps

Outcomes:



Strong Generative AI foundation



Ability to build LLM apps with
APIs, LangChain, and RAG



Portfolio-ready Streamlit
capstone + GitHub repository

Next steps for learners:

- Choose a capstone idea early and iterate weekly
- Keep prompts, experiments, and evaluation notes organized
- Prepare for final demo/viva with a clear story

Thank You

Questions?

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